

Josue Torres-Fonseca

josuextorres@gmail.com | linkedin.com/in/josuetorresfonseca | josuetorresfonseca.com

EDUCATION

Doctor of Philosophy in Computer Science and Engineering University of Michigan, Ann Arbor, MI	Expected May 2028 GPA: 4.0
Master of Science in Computer Science and Engineering University of Michigan, Ann Arbor, MI	Graduated May 2025 GPA: 4.0
Bachelor of Science in Computer Science, Minor: Applied Mathematics Boise State University, Boise, ID	Graduated May 2023 GPA: 3.889, <i>Magna Cum Laude</i>

TECHNICAL SKILLS

Python, LLMs, VLMs, PDDL, PyTorch, NumPy, Pandas, Matplotlib

WORK EXPERIENCE

PhD Candidate, <i>Situated Language and Embodied Dialogue Group</i> University of Michigan, Computer Science and Engineering Division	Aug 2023 – Present
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- Supervised by Dr. Joyce Chai
- Created a safety-centric dataset using PDDL to generate trajectories demonstrating how to complete household tasks while mitigating hazards rendered in the AI2-Thor simulator
- Evaluated 11 state-of-the-art VLMs on hazard recognition via Visual Question Answering (VQA) and safe plan generation as an embodied agent performing household tasks while proactively mitigating hazards in AI2-THOR
- Developed a decoupled multi-agent framework that separated hazard recognition from task execution, resulting in increased safety-planning success rates
- Presented Poster at the Association for Computational Linguistics (ACL) 2026

Undergraduate Researcher, <i>Speech Language and Interactive Machines Group</i> Boise State University, Department of Computer Science	Aug 2021 – Aug 2023
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- Supervised by Dr. Casey Kennington
- Project 2:
 - Designed and built an incremental spoken dialogue system capable of transcribing and understanding speech, detecting, and gaining a concrete understanding of objects, and mapping objects using the Anki Cozmo robot
 - Processed and analyzed spoken interaction data gathered from participants to evaluate the dialogue system
 - Presented at Special Interest Group on Discourse and Dialogue (SIGdial) 2022
- Project 1:
 - Gathered human appraisals and descriptions of robot emotional behaviors from participants
 - Processed and analyzed gathered data
 - Presented at Language Resources and Evaluation Conference (LREC) 2022

Undergraduate Researcher, <i>Language & Intelligence Group</i> Massachusetts Institute of Technology, Department of Electrical Engineering and Computer Science	May 2022 – Nov 2022
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- Supervised by Dr. Jacob Andreas
- Designed and built an interpretable ML model providing neuron annotations using no data
- Presented poster and delivered lightning talk at end of MIT Summer Research Program

Software Engineer Intern Paylocity	May 2021 – Aug 2021
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- Contributed to software development across the full lifecycle as part of a 13-member team

- Worked with product owners and leads to translate use cases into features
- Designed, developed, tested, and deployed new features in project releases
- Applied Agile and Scrum methodologies; gained full-stack experience with JavaScript, React/Redux, C#, .NET, SQL, and Git

Undergraduate Researcher, *Intelligent Security Group*

June 2020 – Aug 2020

Texas State University, Department of Computer Science

- Supervised by Dr. Mina Guirguis; funded by NSF as a Smart & Connected Communities REU site
- Designed and built a game and reward system to secure a simulated water tank using game theory
- Simulated and secured a water tank with check blocks acting as both an intrusion detection and prevention system
- Prepared presentations and report on research findings for REU participants and organizers

TEACHING EXPERIENCE

Tutor, *Operating Systems*

Jan 2023 – May 2023

Boise State University, Department of Computer Science

- Taught students how concepts in class apply to their programming assignments
- Educated students on how concepts taught in class may be used in industry settings
- Assisted students in planning out and debugging their programming assignments

Tutor, *Introduction to Theory of Computation*

Jan 2021 – May 2021

Boise State University, Department of Computer Science

- Assisted students by guiding them through their assignments
- Facilitated individual and group tutoring sessions

Tutor, *Data Structures*

Aug 2020 – Dec 2020

Boise State University, Department of Computer Science

- Worked with students to enhance their understanding of topics discussed in class
- Guided students through debugging their code and understanding how their code worked
- Communicated and effectively explained the purpose of data structures and how they worked

Lab Assistant, *Microprocessors*

Aug 2020 – Dec 2020

Boise State University, Department of Electrical and Computer Engineering

- Guided students through lab work and debugging hardware and software issues
- Helped students understand concepts related to the labs and how the labs demonstrated these concepts
- Assisted professor in grading papers

CONFERENCES & PRESENTATIONS

SafetyALFRED: Evaluating Safety-Conscious Planning of Multimodal Large Language Models

July 2026

Conference Paper

- **Findings of the Association for Computational Linguistics: ACL 2026**, San Diego, CA, USA
- Josue Torres-Fonseca, Naihao Deng, Yinpei Dai, Shane Storks, Yichi Zhang, Rada Mihalcea, Casey Kennington, Joyce Chai. 2026. *SafetyALFRED: Evaluating Safety-Conscious Planning of Multimodal Large Language Models. In Findings of the Association for Computational Linguistics: ACL 2026, pages 37190-37211, San Diego, California, United States. Association for Computational Linguistics.*

The Mechanistic Emergence of Symbol Grounding in Language Models

July 2026

Conference Paper

- **International Conference on Machine Learning 2026**, Seoul, South Korea

- Shuyu Wu, Ziqiao Ma, Xiaoxi Luo, Yidong Huang, Josue Torres-Fonseca, Freda Shi, Joyce Chai. 2026. *The Mechanistic Emergence of Symbol Grounding in Language Models*. In *Proceedings of the International Conference on Machine Learning 2026*, Seoul, South Korea.

rrSDS 2.0: Incremental, Modular, Distributed, Multimodal Spoken Dialogue with Robotic Platforms Aug 2025
Conference Paper

- **Special Interest Group on Discourse and Dialogue**, Avignon, France
- Anna Manaseryan, Porter Rigby, Brooke Matthews, Catherine Henry, Josue Torres-Fonseca, Ryan Whetten, Enoch Levandovsky, and Casey Kennington. 2025. *rrSDS 2.0: Incremental, Modular, Distributed, Multimodal Spoken Dialogue with Robotic Platforms*. In *Proceedings of the 26th Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 637–640, Avignon, France. Association for Computational Linguistics.

Symbol and Communicative Grounding through Object Permanence with a Mobile Robot **Sept 2022**
Conference Presentation and Paper

- **Special Interest Group on Discourse and Dialogue**, Edinburgh, Scotland
- Josue Torres-Fonseca, Catherine Henry, and Casey Kennington. 2022. *Symbol and Communicative Grounding through Object Permanence with a Mobile Robot*. In *Proceedings of the 23rd Annual Meeting of the Special Interest Group on Discourse and Dialogue*, pages 124–134, Edinburgh, UK. Association for Computational Linguistics.

HADREB: Human Appraisals and (English) Descriptions of Robot Emotional Behaviors **June 2022**
Video Presentation and Paper

- **Language Resources and Evaluation Conference**, Marseille, France
- Josue Torres-Fonseca and Casey Kennington. 2022. *HADREB: Human Appraisals and (English) Descriptions of Robot Emotional Behaviors*. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 5739–5748, Marseille, France. European Language Resources Association.

HONORS & AWARDS

Gemini Academic Program Award	(\$10,000)	Spring 2026 – Spring 2027
NSF Graduate Research Fellowship	(\$159,000)	Fall 2023 – Spring 2028
Dean's List		Fall 2018 – Spring 2023
Boise State University Top Ten Scholar		Spring 2023
Barry M. Goldwater Scholarship	(\$7,500)	Fall 2022 – Spring 2023
Federal Employee Education & Assistance Fund Scholarship	(\$1,000)	Fall 2018

ORGANIZATIONS

Treasurer, Computer Science and Engineering Graduate Students **Fall 2024 – Spring 2025**

- Managed organization funds
- Assisted in organizing events

President, Artificial Intelligence Club **Fall 2024 – Spring 2025**

- Facilitated a variety of professional development and technical development workshops about Python, machine learning, and data science

President, Competitive Cybersecurity and Cyber-Physical Club **Fall 2019 – Spring 2022**

- Participated in CyberForce Competition at INL (46th out of 103 teams, Fall 2019)
- Presented on cybersecurity topics including data scraping and reverse engineering with Ghidra
- Led Capture the Flag activities to build foundational cybersecurity knowledge among members